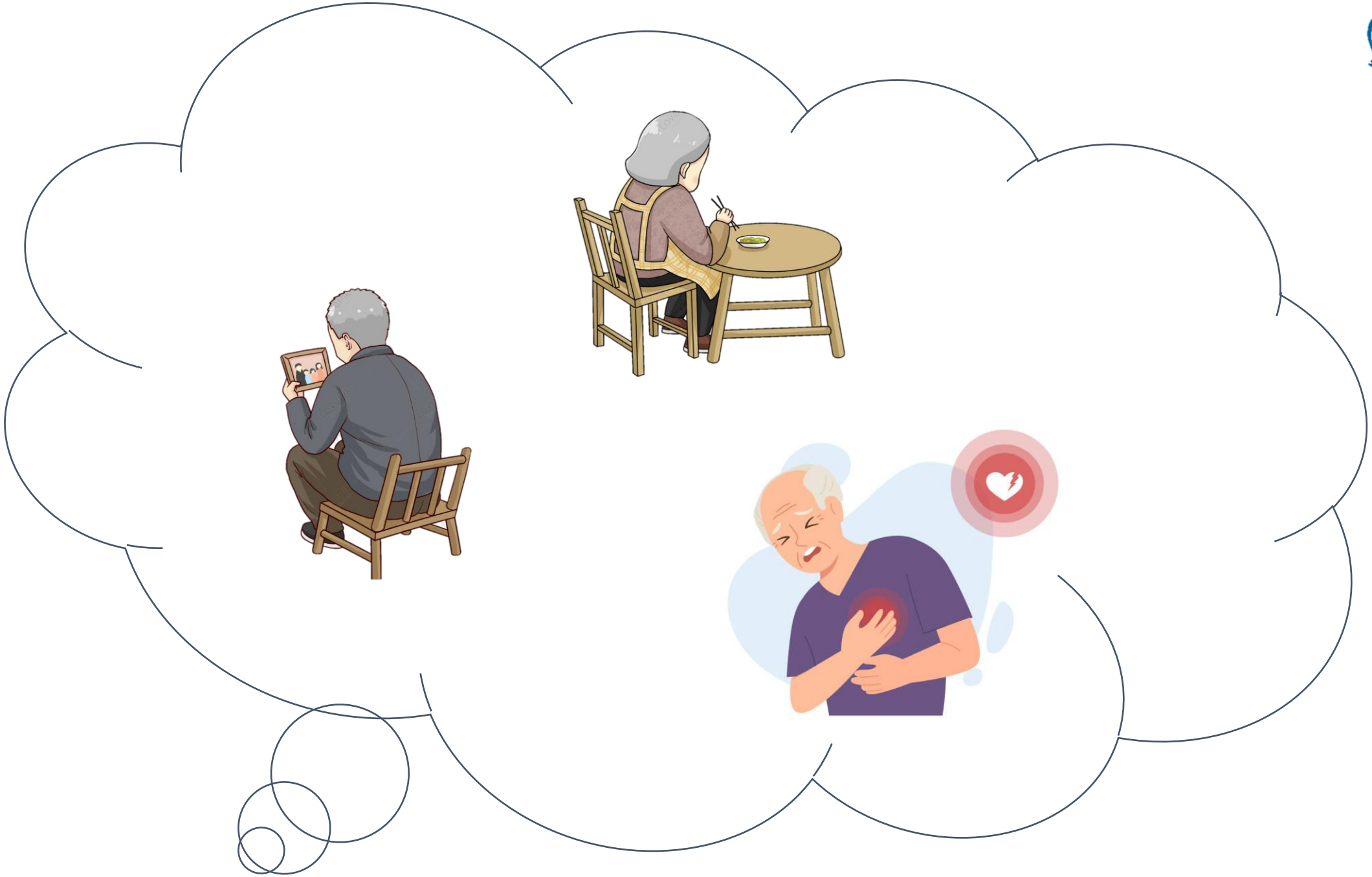
G-OLD x UAiTED
University Alliance in Talent Education Development

LOOK AFTER

The smart-home caregiver





PROBLEMS



COSTLY HOMECARE SERVICES AND DEVICES

SHORTAGE OF HEALTHCARE WORKERS





LACK OF REAL-TIME HEALTH DATA

COSTLY HOMECARE SERVICES



- Nursing home care monthly costs per patient were estimated to be US$1000(NT$30,000), which the major component of costs incurred by families (at least 78% of total family costs).
- The higher costs were encountered for caregiving at home when the patients had severe dependence.

COSTLY HOMECARE SERVICES



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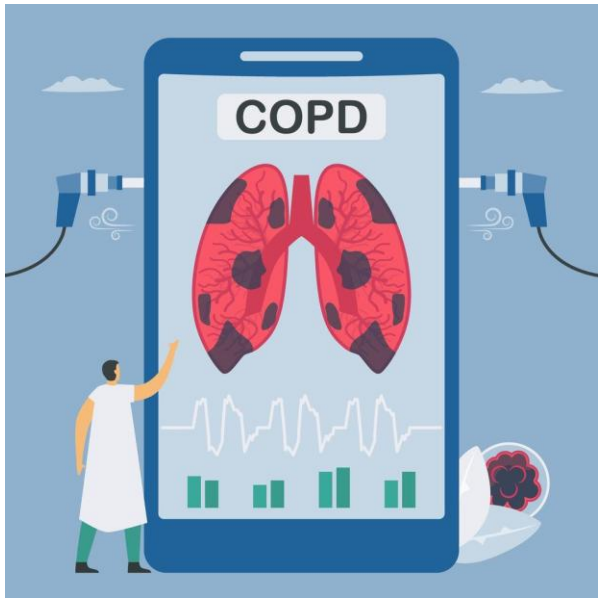
COSTLY HOMECARE SERVICES



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REAL-TIME MONITORING

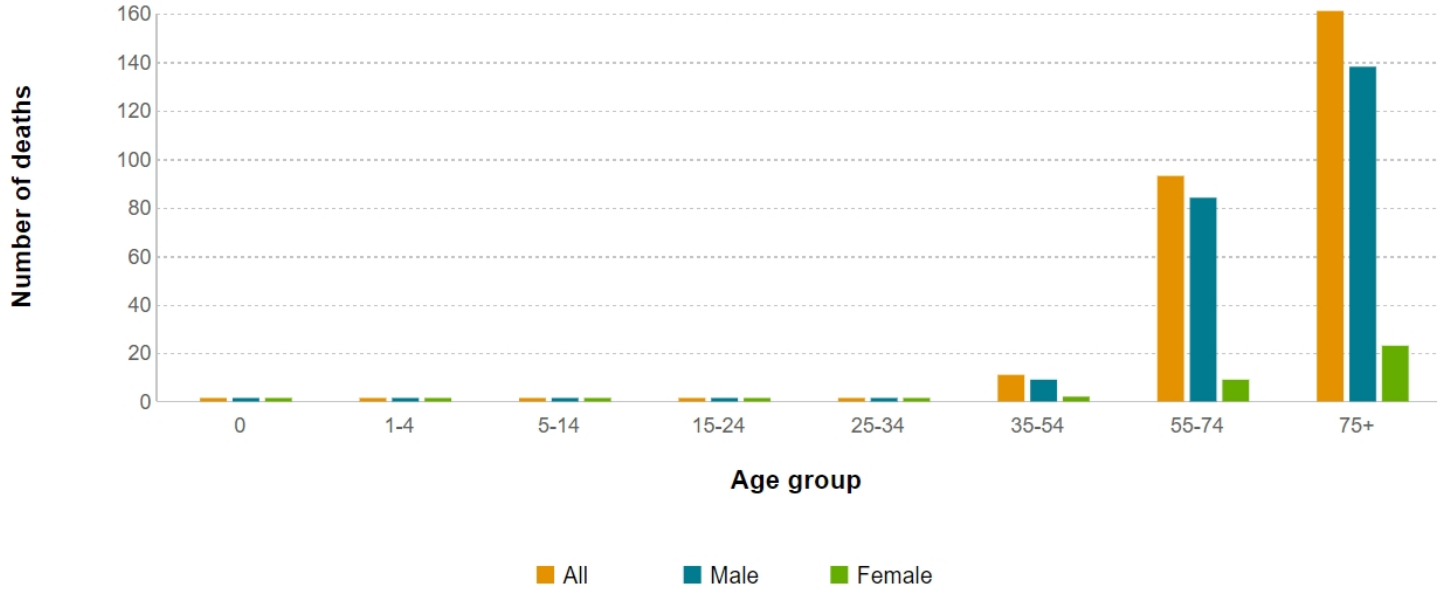
Chronic Obstructive Pulmonary Disease (COPD)



According to WHO, COPD is the

3rd leading cause of death worldwide

Chronic obstructive pulmonary disease (Singapore, 2020)



Complications:

- Malnutrition and Sarcopenia
- Cardiovascular Disease
- Lung Cancer
- OSA
- Osteoporosis

Reference: Chronic obstructive pulmonary disease. Who.int. Published 2020. https://platform.who.int/mortality/themes/theme-details/topics/indicator-groups/indicator-group-details/MDB/chronic-obstructive-pulmonary-disease

FALL DETECTION

Complication:

- Hip fractures
- Prolonged immobility
- Death

衛生福利部
Ministry of Health and Welfare
促進全民健康與福祉

請輸入關鍵字 進階

熱門關鍵字： COVID-19 長期照顧 防疫補償 十大死因

本部簡介 最新消息 便民服務 法令規章 衛教視窗 重大事件 本部各單位及所屬機關

108年衛生福利部新聞

9月新聞

首頁 / 最新消息 / 焦點新聞 / 108年衛生福利部新聞 / 9月新聞

每6人就有1位老人曾跌倒 國健署傳授防跌妙招

- 資料來源：國民健康署
- 建檔日期：108-09-24
- 更新時間：112-07-20

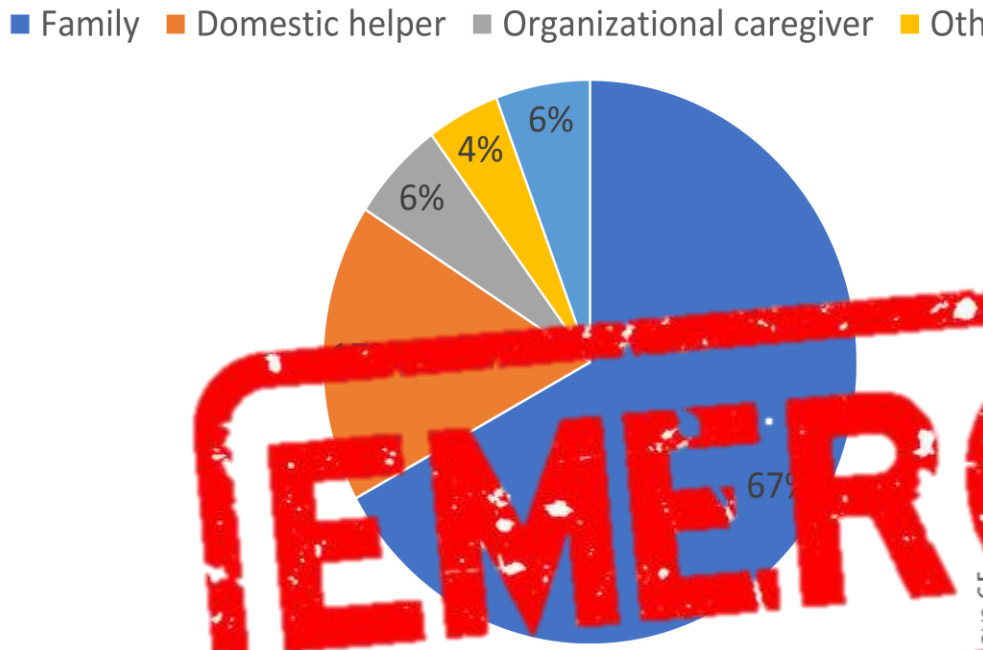
107年死因統計中，跌倒高居65歲以上事故傷害死亡原因第二位(每十萬人25.7人)。根據國民健康署106年「國民健康訪問調查」3,280位65歲以上老人中，自述過去一年曾跌倒者有495人(15.5%)，也就是說每6個老人，就有1位在一年內有跌倒的經驗。嚴重跌倒可能會造成長期臥床甚至死亡，國民健康署呼籲長者，防跌要從生活小細節做起，切勿輕忽跌倒的傷害，規律運動、維持居家環境及用藥安全，都是防跌的好方法。

超過6成長者跌傷後 仍未採取防跌措施

國民健康署106年「國民健康訪問調查」結果顯示，65歲以上老人自述過去一年曾跌傷且有就醫者為264人(8.4%)，跌傷就醫後僅約2成長者會注意提升居家安全，逾6成長者跌傷後未採取任何防跌措施(圖一)，進一步分析長者跌傷地點，室內發生跌傷的第一位為臥室(35.6%)，第二位為客廳(30.8%)，第三位為浴室(17%)(圖二);室外發生跌傷的地點第一位為街道或路上/路邊(62.9%)，第二位為公園/運動場或溜冰場(10.4%)，第三位為菜園/農地(9.4%)(圖三)；另外，從長者跌傷的原因來看，第一位為滑倒或絆倒(43.2%)，第二位為失去平衡或頭昏(37.1%)，第三位為碰到或撞上物品或人(11.7%)(圖四)。一般而言，長者會因為上述原因跌傷，可能和身體功能下降(視力、聽力、肌力、平衡感)、生病、使用鎮定藥物或合併多種用藥等有關。

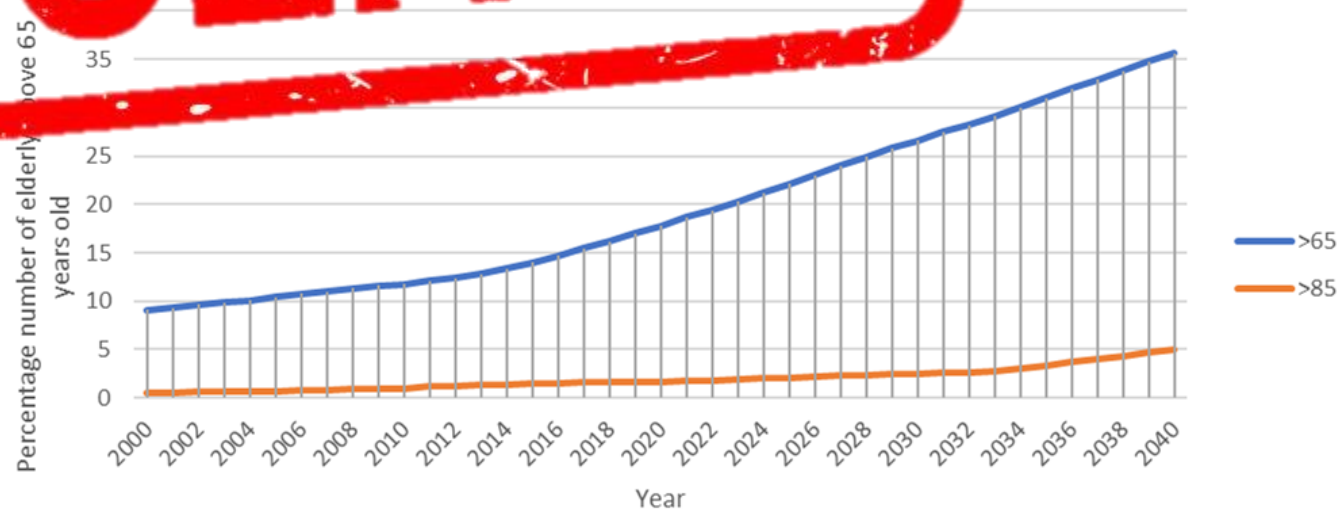
https://www.nytimes.com/2017/01/16/well/live/getting-older-sleeping-less.html#:~:text=The%20causes%20of%20insomnia%20are,including%20symptoms%20of%20cognitive%20loss.

Elders' Assistive Living in Taiwan



EMERGENCY

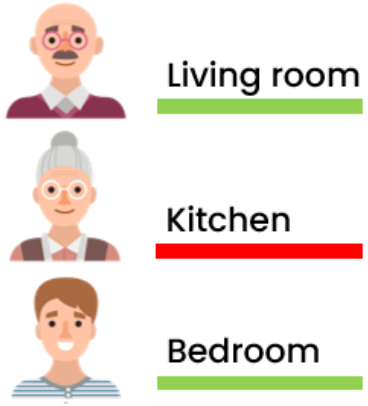
Reference: iHealth. 台灣老人 … 比例?. iHealth 健康報報. Published December 25, 2018. https://www.ihealth.com.tw/article/%E5%8F%B0%E7%81%A3%E8%80%81%E4%BA%BA%E5%BF%AB%E6%A8%82%E6%8C%87%E6%95%B8/



Reference: 高齡人口占比趨勢 - 人口推估統計查詢系統. Ndc.gov.tw. Published 2022. https://pop-proj.ndc.gov.tw/chart.aspx?c=10&uid=66&pid=60



Activity Recognition



Fall Detection





LOOK AFTER

Non-intrusive health monitoring device

Vital Sign



15BPM

Sleep Monitoring



APP PROTOTYPE

Home



GG-OLD

FAVORITES ALL

Grandpa

Tainan

Grandma

Taipei

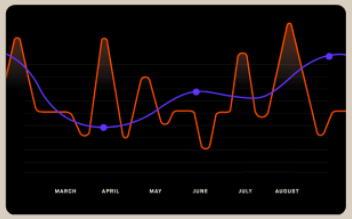
HOME STATISTIC ME

Statistics

Statistics

Heart Beat Rate

Day Week Month



Unusual Heart Beat Detected

Recommended Activities

1. Seek immediate medical help: If you are experiencing an unstable heart rate or any cardiac symptoms such as chest pain, dizziness, shortness of breath, or fainting, call emergency services or visit the nearest emergency room without delay
2. Maintain a healthy lifestyle: Adopt a heart-healthy lifestyle to support your overall cardiovascular health. This includes:
   - Eating a balanced diet rich in fruits, vegetables, whole grains, lean proteins, and healthy fats.
   - Engaging in regular physical activity as recommended by your healthcare provider.
   - Avoiding tobacco products and limiting alcohol consumption.
   - Managing stress through relaxation techniques, such as deep breathing exercises, meditation, or yoga.

HOME STATISTIC ME

Click In Device

Grandpa's Device

Current Activity

EATING

Last Update : 21:30



Activity Status for 5/6/2023

Sleep Tracking

★REM : 1 Hour 45 Minutes

★CORE : 4 Hours 48 Minutes

★DEEP : 45 Minutes

★AWAKE : 28 Minutes

HOME STATISTIC ME

Real Time Monitoring

Grandpa's Device

Vital Signs Monitoring: Status:



BPM: 80



RESP: 15



mmHg: 90



Walk Motion

Abnormalities Detected : 0

1255 Steps / 0.994 km



Breathing Motion

COPD : Not Detected

Asthma : Not Detected

Last Update : 21:30

HOME STATISTIC ME

Notification Page

Notifications

Messages

Grandma's Device 21:33

Motion was detected at 21:33:09.

Grandpa's Device 20:30

Motion was detected at 20:30:59.

New Device Added 20:25

LF-1000 Wi-Fi Sensing was added

HOME STATISTIC ME

Channel State Information (CSI)

HOW IT WORKS

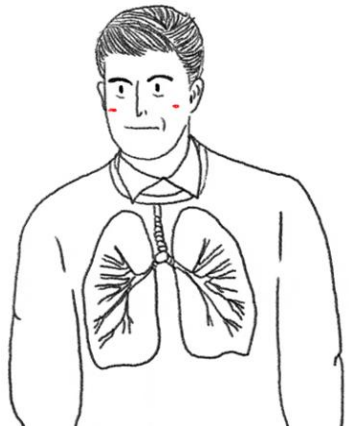


LOOK AFTER

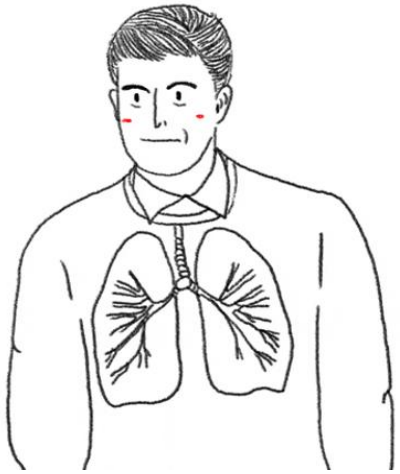




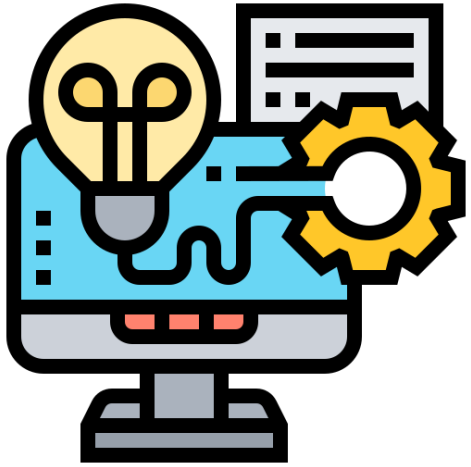




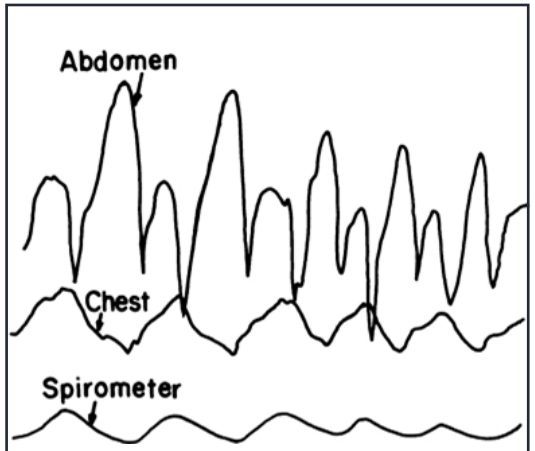
HOW IT WORKS



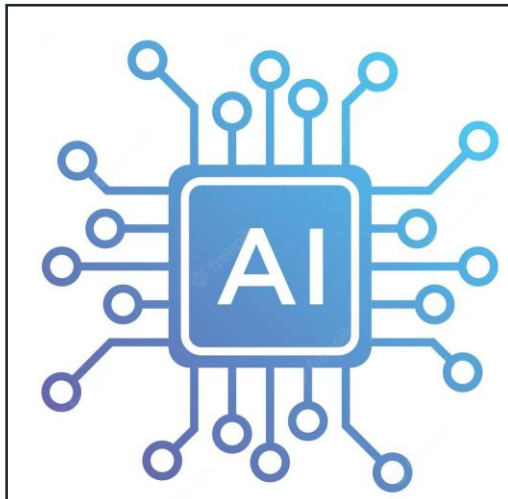
Collect Datasets



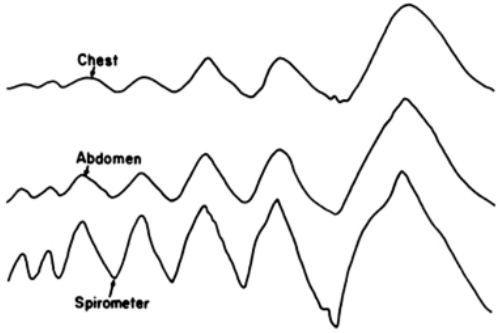
Machine Learning



Anomaly detection



Elaborate using AI



Normal breathing pattern

EXPECTED BENEFITS

CHEAPER	Affordable health monitoring
BETTER	Delivering real-time health insights
FASTE R	Initiating immediate emergency response
ACCESSIBLE	Facilitation of remote patient monitoring
SPECIALIZED	Enabling proactive prevention through early detection

Cost Per Unit



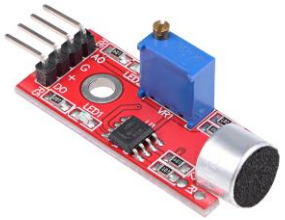
Product Shell (3D Printing)

$10



Grove Ultrasonic Ranger V2

$3



HW-484(Sound Sensor)

$2

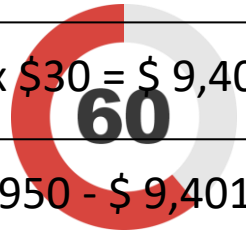


ESP32-CAM WIFI Bluetooth Board &
OV2640 Camera Module with 2.4 Antenna CH340G

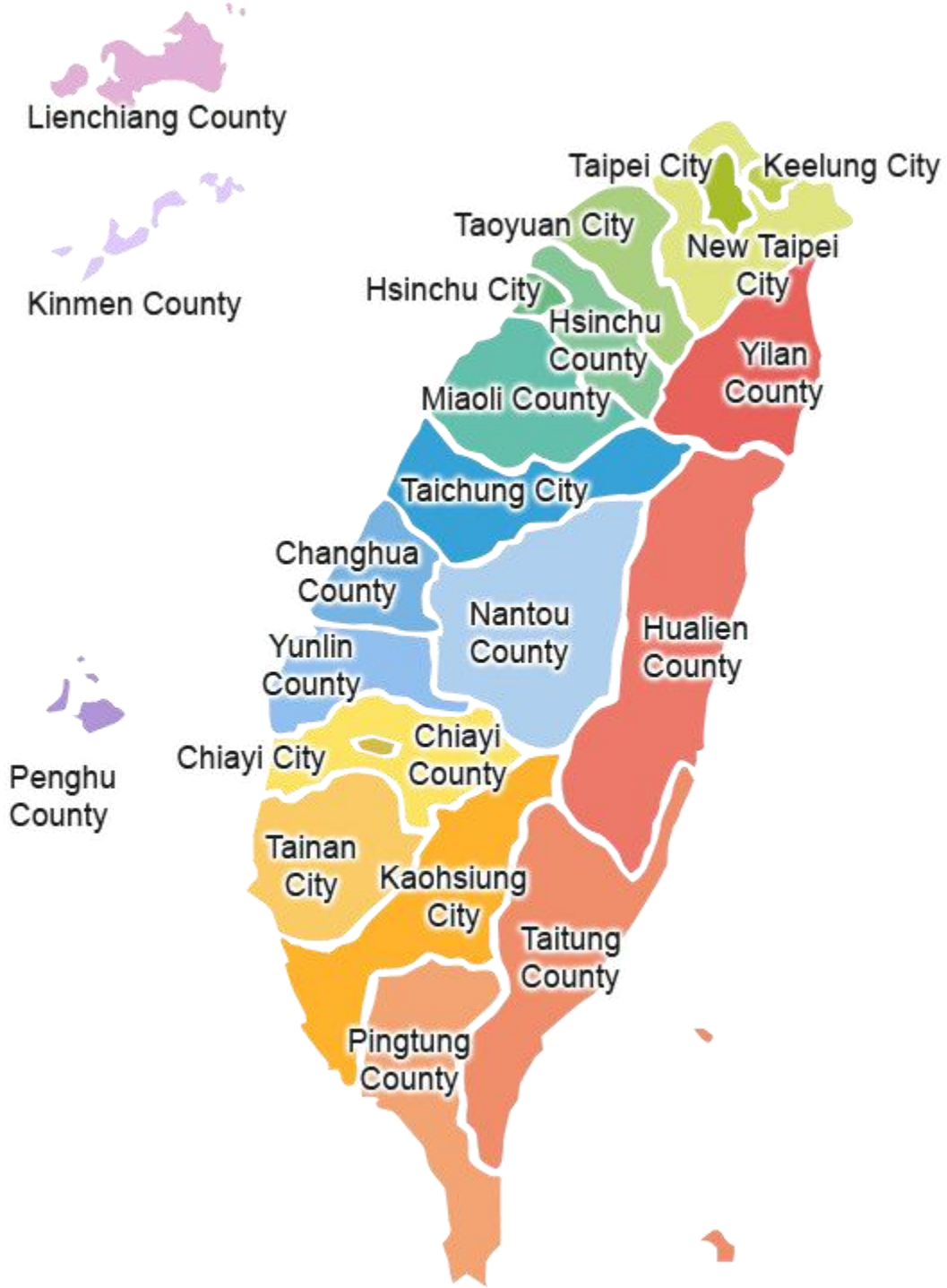
$12



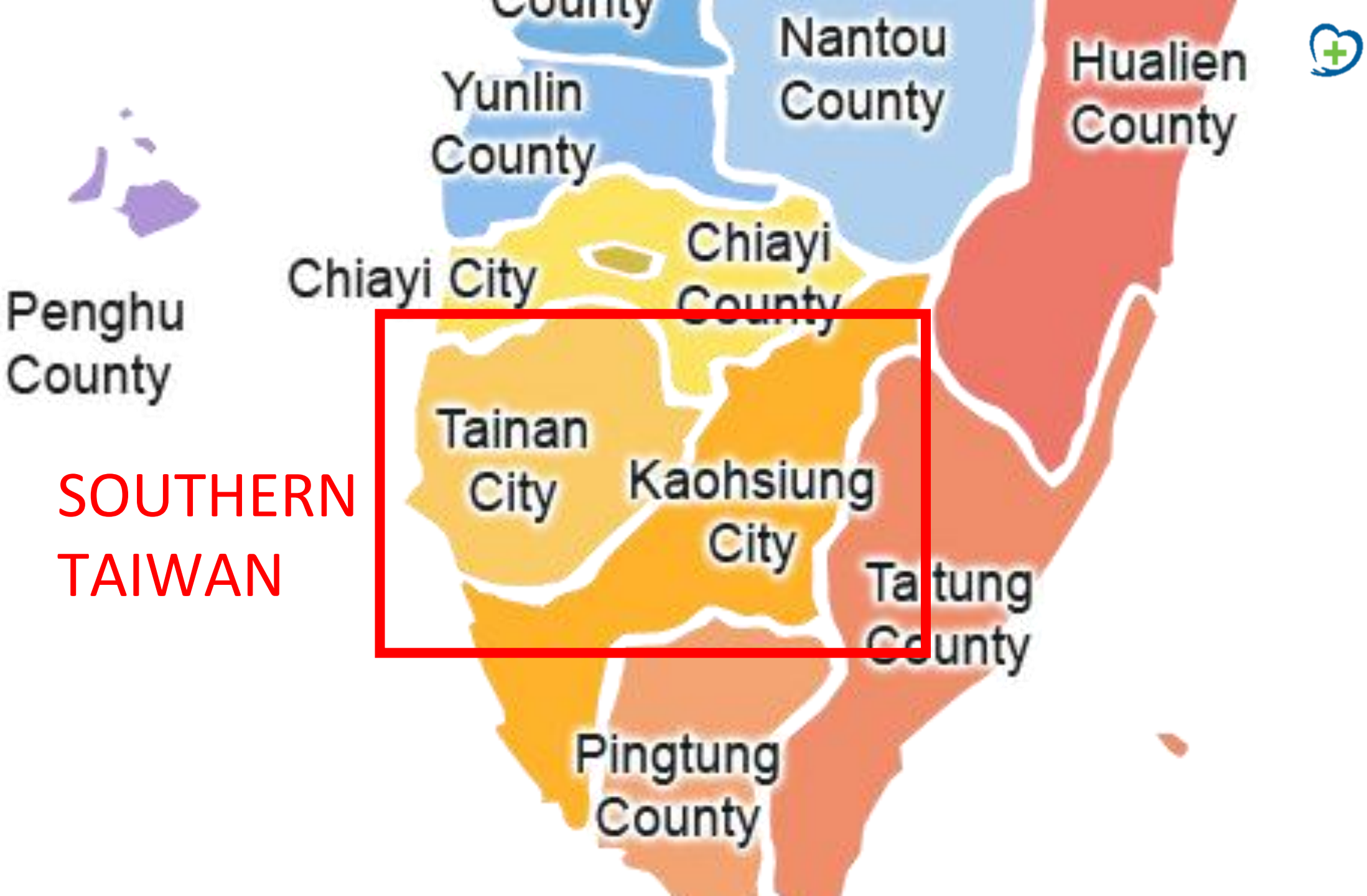
Revenue	313,399 x $50 = $ 15,699,950
Cost	313,399 x $30 = $ 9,401,970
Profit	$ 15,699,950 - $ 9,401,970 = $ 6,297,980 (NT$ 198,222,623)



BEACH-HEAD MARKET



BEACH-HEAD MARKET



BEACH-HEAD MARKET

3 Tainan City

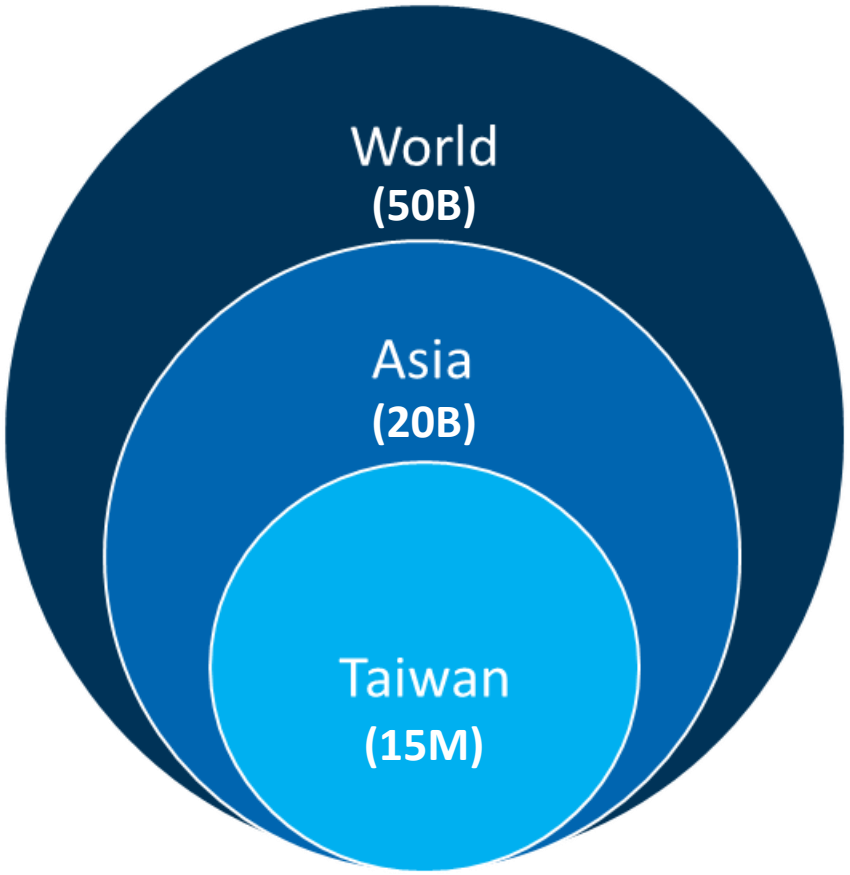
- The ratio of people over 65 accounted for around 4M people
- Accounted for 48.44% of assistance provided for the elder



2 Kaohsiung City

- The ratio of people over 65 accounted for 18.33%
- Accounted for 44.06% assistance provided for the elder

Total Market Size



In the year 2022, there are 313,399 applicants on Taiwan MOHW LTC2.0 Program



x $50 = 15,699,950 USD

(NT$494,171,626)

COMPETITOR ANALYSIS

	Look After	*Google Nest Mini*	*BioIntelliSense*	*TP-Link C200*
Product				
Price	NT$1500=US$50	NT$1785 = USD$57	NT$12900 = USD$412	NT$849 = USD$27
Real-time vital sign monitoring	✓	✓		✓
Circular Economy	✓		✓	
Biodegradable materials	✓	✓		
Elimination of Wearables	✓			

Project Timeline



Start Research
Wi-Fi Sensing
Technology



Regulatory
& Certification



Product
Development &
Design



Done Research

Look For Partners



3rd Funding
USD 100,000

Launch

Mass Production

Done Design &
Development

Alpha & Beta Test

Application
Development

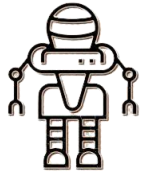
2nd Funding
USD 30,000

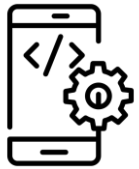
Prototyping

Pre Seed Funding
USD 5,000

Pilot Test
Get Feedback and
Patient
Data/Information



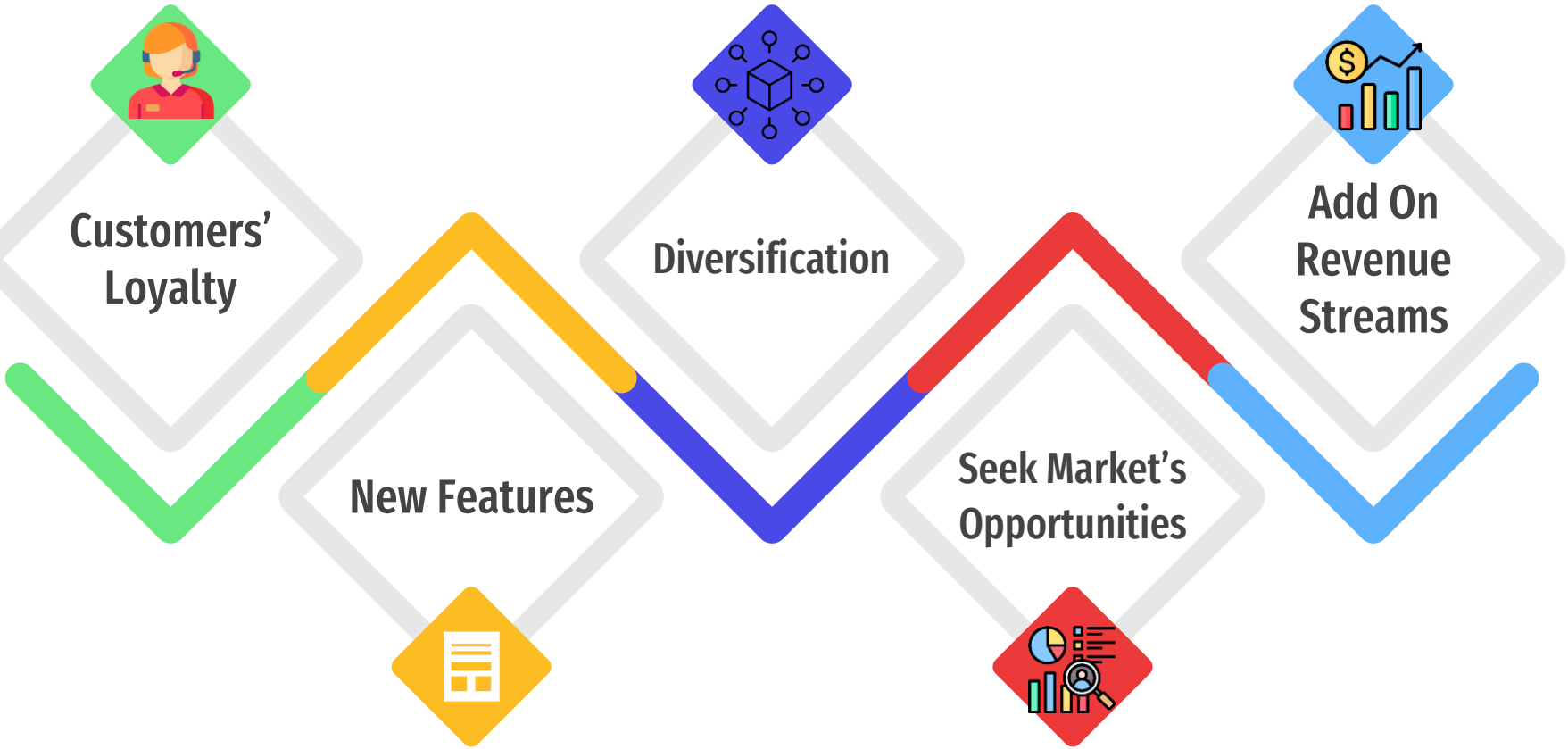








OUR MARKET GOALS



POTENTIAL MARKET EXPAND

Taiwan



Stage 1

People over 65: 16.91% (2022)

Singapore



Stage 2

People over 65: 16.6% (2022)

Other Parts of Southeast Asia



Stage 3

People over 65: 9.5% (2022)

Potential Partners:



行政院 Executive Yuan

RafflesMedicalGroup

THOMSON MEDICAL

DKSH

MGP PT. MEGA GLOBAL PRATAMA The Ultimate Trading Solution

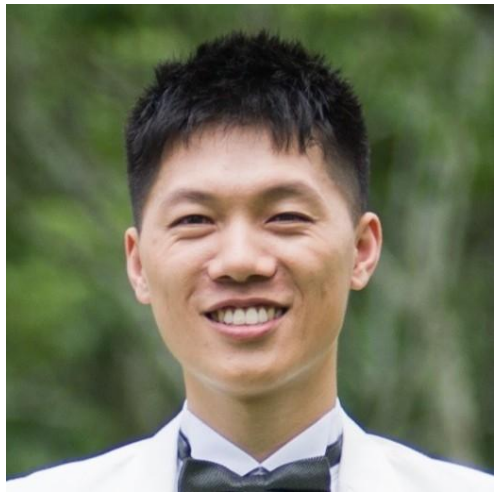
MEDIHUB BETTER HEALTH COMMUNITY

TESTIMONIALS



Dr. Jun-Lian, Guo
Resident physician, Department of Neurology, Tri-service general hospital, Taipei

"It will make society become safer for those elderly population."



Dr. Ke-Wei, Chen
Attending physician, Department of Family medicine, Kaohsiung Armed Forces General Hospital

"This is a good idea, even in nursing homes, they could not monitor them continuously."



Dr. Cheok Kang Ni
General Doctor of Klinik Ke
Tebrau, Johor, Malaysia

"It reduces the burden of caregivers as it does not need to look into th elderly all the time"

TEAM MEMBERS

G-OLD

NATALIE TANJAYA

Biomedical Engineering, NCKU Taiwan



Chief Executive Officer (CEO)
Responsible for setting the overall direction and strategy.

LAM NGUYEN NGAN HA

Biomedical Engineering, NCKU Taiwan



Chief Product Officer (CPO)
Overseeing the company's product or service development

YEOH ZHAO WEI

Software Engineering, UTAR Malaysia



Chief Technology Officer (CTO)
Responsible for managing the company's finances

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

1931 國立成功大學 National Cheng Kung University

UTAR UNIVERSITI TUNKU ABDUL RAHMAN



SI POH YI

Civil Engineering, NTU Singapore

Chief Marketing Officer (CMO)
Developing and executing the company's marketing strategy



CHAI JIN XIE

Role: Develop and manage technology resources

Achievement: Sucess in engineering and tech competition such as top 3 in EIC 2022 and finalist in both ongoing NTU Port63 Challenge and EIC 2023

Look After G-old

"Prevention is one of the few known ways to reduce demand fo health and aged care services."

Julie Bishop

(Australian Former Minister for Foreign Affairs, 1956)





Special Thanks:
Mr. Paul Singh Gill (Team Mentor)
i-Train

APPENDIX

Financial Year 2024

	Project	Content	Cost Estimate (USD)
1	**Early Product Development**	**Wifi Sensor Design, AI System Establishment AND Wifi Sensor Settings with the Development Board**	**$2,000**
		Wifi Sensor Testing AND Prototype Design	**$3,000**
2	**Expert Consulting**	**Business Model, Regulations and Patent Experts**	**$7,000**
3	**Patent Layout**	**PA AND PCT**	**$15,000**
		Invention Patent, Utility Model Patent, Design Patent	**$900**
4	**Salary**	**Total Salary For 2024**	**$12,000**
5	**Operation in Taiwan**	**Business Registration**	**$38**
		Accounting and Audit (Every 14 Months)	**$1,400**
		Laboratory and Office Rental (Per Year)	**$25,000**
		Labour and Health Insurance (Per Year)	**$1000**
		Internet (HiNet) (Per Year)	**$600**
		Utility Bill (Per Year)	**$1,200**
6	**Regulatory Certification**	**(Taiwan) BSMI + NCC: certificate (per year)**	**$800**
7	**Others**	**Consumables and Transportation**	**$1,500**
Total			**$ 69,138** **NT$ 2,166,079.71**

Financial Year 2025

	Project	Content	Cost Estimate (USD)
1	**Product Development**	**Wifi Sensor Alpha & Beta Testing**	**$5,000**
		Application Development AND Design	**$1,000**
		Adjustable Element Design AND Mass Production Test	**$5,000**
2	**Expert Consulting**	**Wifi Technology AND Device Testing Experts**	**$3,000**
3	**Salary**	**Total Salary For 2025**	**$60,000**
4	**Operation in Taiwan**	**Accounting and Audit (Every 14 Months)**	**$1,400**
		Laboratory and Office Rental (Per Year)	**$25,000**
		Labour and Health Insurance (Per Year)	**$1000**
		Internet (HiNet) (Per Year)	**$600**
		Utility Bill (Per Year)	**$1,200**
5	**Regulatory Certification**	**(Taiwan) BSMI + NCC: certificate (per year)**	**$800**
6	**Others**	**Consumables and Transportation**	**$3,500**
Total			**$ 107,500** **NT$ 3,367,953.50**

Financial Year 2026

	Project	Content	Cost Estimate (USD)
1	**Product Development**	Mass Production	$20,000
		Research AND Develop New Functions	$5,000
2	Expert Consulting	Marketing Experts	$1,000
4	Salary	Total Salary For 2026	$90,000
5	Operation in Taiwan	Business Registration	$38
		Accounting and Audit (Every 14 Months)	$1,400
		Laboratory and Office Rental (Per Year)	$25,000
		Labour and Health Insurance (Per Year)	$1000
		Internet (HiNet) (Per Year)	$600
		Utility Bill (Per Year)	$1,200
6	Regulatory Certification	(Taiwan) BSMI + NCC: Laboratory test + certificate (per year), (Singapore) IMDA: Laboratory test + certificate (per year)	$900
7	Others	Consumables and Transportation	$3,500
Total			**$ 146,138** **NT$ 4,578,474.31**

	2024	2025 - 2026	2027- 2030
	Short-term	Mid-term	Long-term
MILESTONES	Marketing Site Launch BETA Launch Launch	Obtained 5% market in Taiwan – around 200,000 Users Approach international market (Singapore and Malaysia) with 100,000 Users	Achieve 1 million users in TW and 1 million users in Singapore and Malaysia Expand the new potential customers in Indonesia and Vietnam Retain old customers and reach new customers in other areas
PRODUCT	Customer Testing Mobile App & Website Feedback	Improve user onboarding Usage Analytics System improvement Maintenance and update system	Improve service performance Introduce new features 3rd party integration Customer care and after-sales service
SALES	Create Bottom Up Start-up Strategy Analyze sales trends Competitor Analysis	Competitor Risk analysis (second phase) Connect with distribution networks	Profit opportunities Engage key vendors Prospect Leads Negotiate + Deal
MARKETING	Lead Generation strategies Public Relations Set-up Google Analytics	Partnership Monthly webinar	Loyalty campaign Outreach program Schedule focus group Diversification
OPERATION	Roles Assignment Matrix Create Expense Budget Functional Needs	Round funding raising Market sale reporting Identify company values Expand company size	Cost and complexity drivers Streamline financial reporting

Key Partners

Healthcare Providers & Retailers
Technology Suppliers, Manufacturers, & Distributors
Government Agencies & Industry Players

Key Resources

WiFi Sensing Technology & Infrastructure
Funds
Data Storage & Analysis Capabilities
R&D Team
Partnerships & Collaborations
Regulatory Approvals
Brand Reputation & Customer's Loyalty
Customer Support Team

Customer Segments

Elderly with Respiratory Diseases
Caregivers of Elderly
Healthcare Institutions & Professionals
Smart Home Devices Enthusiasts or Early Adopters

Customer Relationships

Continuous Customer Support
Regular Software Updates & Improvements
Survey & Feedback Gathering
After-sales Support
Warranty
Loyalty Campaign

Key Activities

Short Term:

- Product Development
- Market Research
- Supply Chain & Partnerships
- Pre-launch Marketing & Promotion
- Public Relations

Mid Term:

- Expansion in Taiwan & Approach International Market
- Examine User's Usage & Improve Their Experience
- Expand Company Size & Tailor Future Roadmap
- Loyalty Campaign

Long Term:

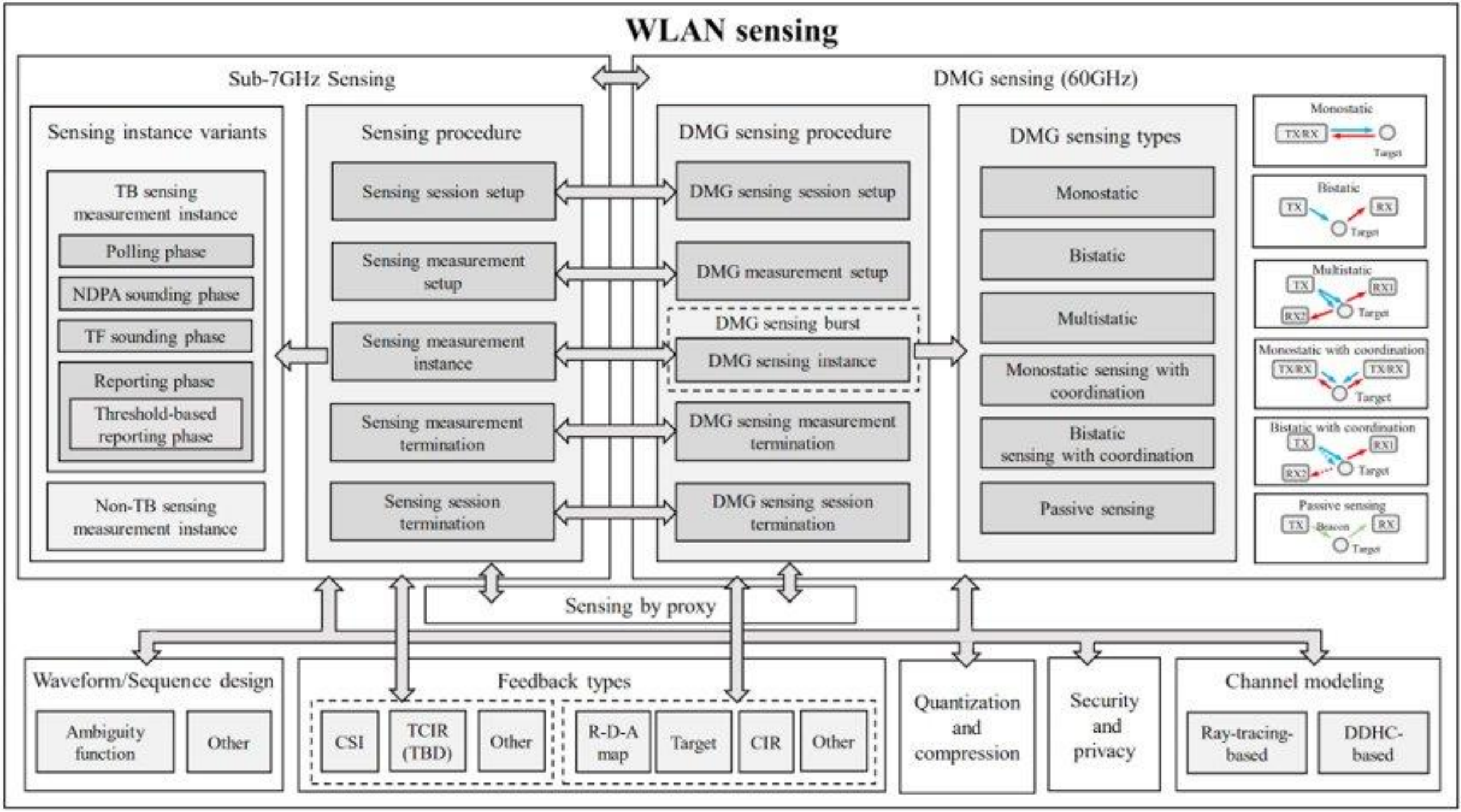
- Expansion in other countries
- Introduce New Features
- Diversification
- Third Party Integration

Channels

Website, App & e-commerce platforms
Partnerships with healthcare institutions for distribution
Collaboration with smart home device manufacturers for bundled offerings
Healthcare seminars & conferences
Social media
Retail Partners

Key Value Propositions

Non-Intrusive Monitoring
Early Detection & Intervention
Real-time Personalized Care & Health Insights
User-friendly, Easy Installation & Affordable


WLAN sensing
Sub-7GHz Sensing
DMG sensing (60GHz)
Sensing instance variants
TB sensing measurement instance
Polling phase
NDPA sounding phase
TF sounding phase
Reporting phase
Threshold-based reporting phase
Non-TB sensing measurement instance
Sensing procedure
Sensing session setup
Sensing measurement setup
Sensing measurement instance
Sensing measurement termination
Sensing session termination
DMG sensing procedure
DMG sensing session setup
DMG measurement setup
DMG sensing burst
DMG sensing instance
DMG sensing measurement termination
DMG sensing session termination
DMG sensing types
Monostatic
Bistatic
Multistatic
Monostatic sensing with coordination
Bistatic sensing with coordination
Passive sensing
Monostatic
TX/RX
Target
Bistatic
TX
RX
Target
Multistatic
TX
RX1
RX2
Target
Monostatic with coordination
TX/RX
TX/RX
Target
Bistatic with coordination
TX
RX1
RX2
Target
Passive sensing
TX
Beacon
RX
Target
Sensing by proxy
Waveform/Sequence design
Ambiguity function
Other
Feedback types
CSI
TCIR (TBD)
Other
R-D-A map
Target
CIR
Other
Quantization and compression
Security and privacy
Channel modeling
Ray-tracing-based
DDHC-based